

New Interface for Musical Expression on a Piece of Wood

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ABSTRACT

A workshop is proposed in which participants will build an electronic circuit on a piece of wood that creates a musical instrument. Creativity and non-conformity are encouraged. Participants will keep their creation with which they can entertain or annoy their family, friends, and colleagues. No prior skills are required; all materials will be provided.

CCS CONCEPTS

• **Hardware** → **Sensors and actuators**; **Sensor applications and deployments**; *Sensor devices and platforms*; **Sound-based input / output**; • **Human-centered computing** → **Gestural input**; **Auditory feedback**; **Activity centered design**; • **Applied computing** → **Media arts**; **Sound and music computing**; • **Social and professional topics** → **Informal education**.

KEYWORDS

Hands On, Maker, Relational Aesthetics, Analog Electronics, Stepped Tone Generator, Atari Punk Console, Forrest Mims, Carpentry

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1 INTRODUCTION

In the early days of amateur electronic experimentation, circuits were built on a piece of wood. Aesthetics ranged from shiny brass hardware and carefully finished and polished wood [Armstrong 2024] to scraps of wood and simple nails [Miclan 2019]. Although printed circuit boards (PCBs) have largely replaced this circuit construction method, there is perhaps something lost, especially to beginners, when the connections between components are almost magically accomplished by the PCB's traces. In this workshop, by building a sound making circuit on a piece of wood, participants will gain a better understanding of how electronics components are connected in order to produce the desired circuit. The popular Atari Punk Console [Mims 2015], first conceived of as the Stepped Tone Generator by Forrest Mims, is a whimsical sound producing device which can be used by anyone to entertain themselves or annoy

others. Finally, most people attending SIGGRAPH spend most of their day sitting at a computer, and the opportunity to bang nails into a piece of wood and route a circuit in any way they wish is a welcome change for many.

2 METHODOLOGY

A brief introduction to the project will include a short history of the device, an explanation of the circuit to be built, and a demonstration of a completed sample device. The difference between the circuit, which must be followed accurately, and the layout, with which the participants will have a great deal of latitude, will be explained. A quick overview of the construction methodology will be provided. Each participant will chose from wooden bases in a variety of shapes, and then will spend some time thinking about how they wish to lay out the circuit on their chosen piece of wood. Double sided tape will be used to anchor the larger (and heavier) components: the loudspeaker and the battery.

The first component, the 556 Dual Timer integrated circuit (IC) will be handed out, and a brief soldering lesson will be provided, after which participants will then solder wires to the pins of the IC. (In fact the IC will be mounted on a small breakout board, making it easier to solder wires to each pin and reducing the likelihood of heat damage to the IC.) Wires of different colors will be used for each node, making the circuit easier to follow for beginners, and easier for the author to identify errors.

Now the fun begins: Participants will hammer nails into their wooden bases for each pin, and will strip and wrap the wires from the IC to the nails. No soldering occurs yet because other components and more wires will be attached. Components will be provided one at a time to avoid confusion, and the necessary connections will be explained. Additional nails will be added as necessary, for example, to connect the 10 microfarad capacitor to the loudspeaker. Each participant will be provided with a paper copy of the circuit and a yellow highlighting pen, which which to mark each connection as it is made, making it easier to see what has been finished and what connections and components have yet to be installed. Step-by-step instructions will also be available for those who prefer.

When all the wires and components have been installed, the connections are soldered, batteries are connected, and joyous sounds ensue.

Step-by-step instructions can be found on the author's website [Shiloh 2024].

3 FINDINGS

The author has conducted many workshops of this sort. Participants report an improved understanding of what electronic components are and how they may be combined to create different circuits, how the same circuit can be built with many different layouts, how light dependent resistors can be used as an effective gestural input device,

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and how different components can control the pitch and duration of tones. Participants report being excited by building a circuit in a most unusual way and pride in building an electronic circuit with no prior experience. Finally, participants enjoy the fun atmosphere and the unique social experience.

4 NEXT STEPS

Participants will be encouraged to hold similar workshops in their communities. A full list of the necessary tools and components, step-by-step instructions, and the author's contact information, will all be available.

5 CITATIONS AND BIBLIOGRAPHIES

REFERENCES

- Armstrong. 2024. Armstrong One-Tube Radio. Retrieved February 18, 2024 from https://www.radiomuseum.org/t/borden_armstrong_one_tube_radio.html
- Miclan. 2019. Wooden Breadboard Circuits.jpg. Retrieved February 18, 2024 from https://commons.wikimedia.org/wiki/File:Wooden_Breadboard_Circuits.jpg
- Forrest M. Mims. 2015. Making Music with an Atari Punk Console. Retrieved February 18, 2024 from <https://www.jameco.com/Jameco/workshop/DIY/atari-punk-console.html>
- Michael Shiloh. 2024. New Interface for Musical Expression on a Piece of Wood. Retrieved February 18, 2024 from <https://michaelshiloh.github.io/nimew/nimew.html>